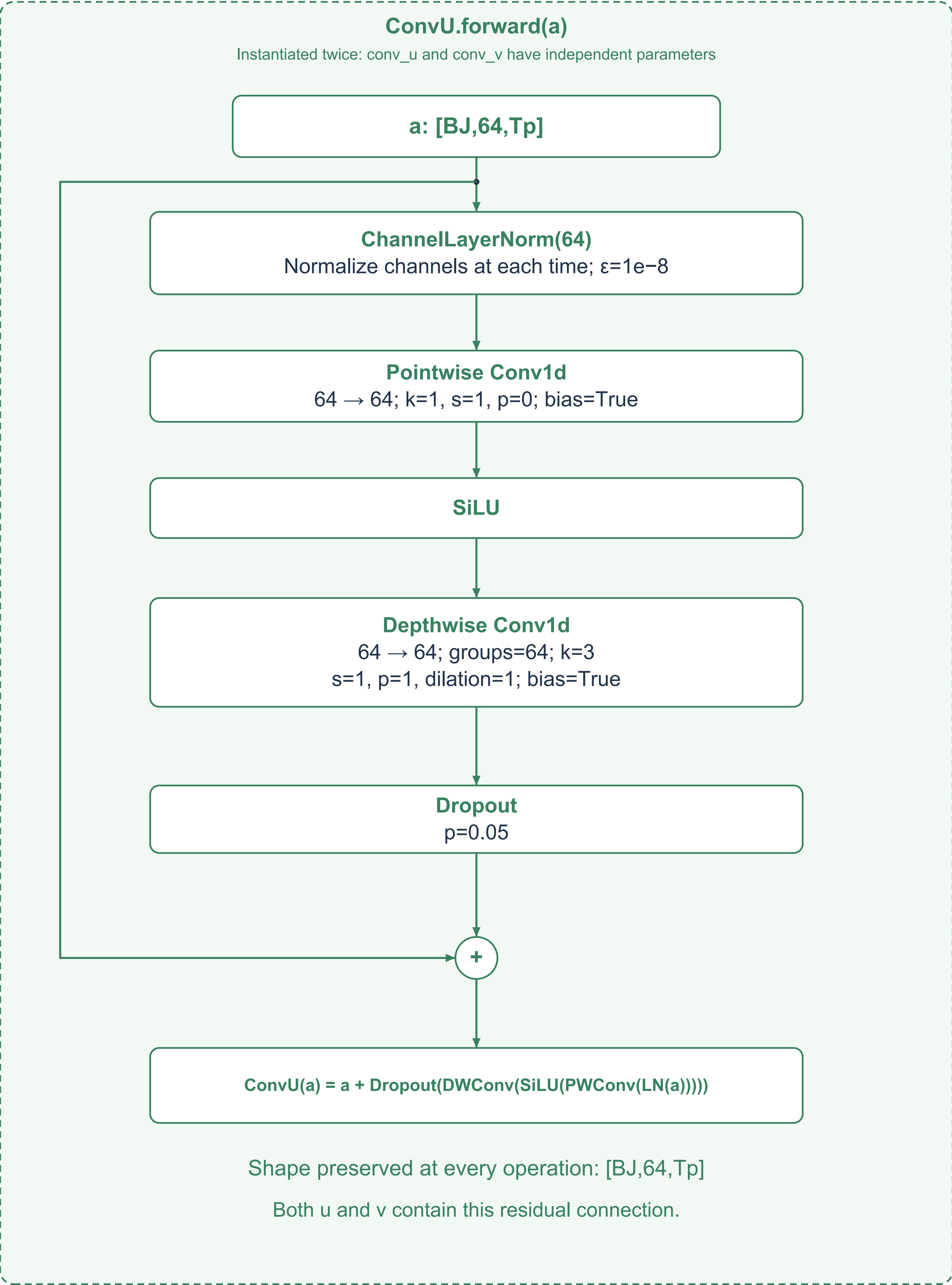
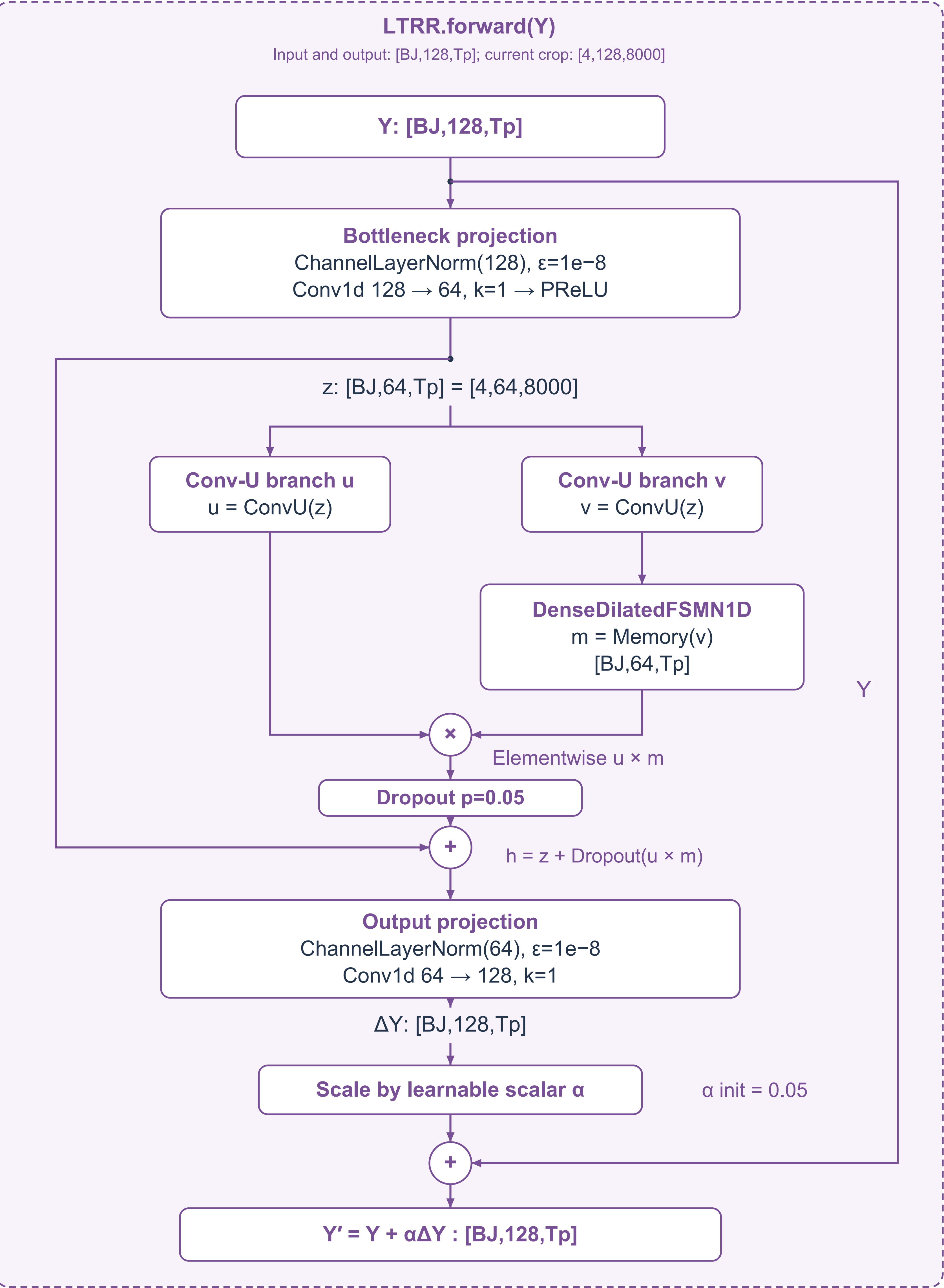


LTRR | residual refinement and the two Conv-U branches

Lightweight Temporal Residual Refinement · original active NoGate operations · only after the final reconstruction stage



The outer bypass is Y; the inner bypass is z; each Conv-U has its own input bypass. Memory itself has no extra residual addition.

PReLU uses one learnable parameter (PyTorch default). LTRR convolutions have bias=True. No speaker-guided gate is instantiated.

Verified parameters: LTRR = 89,031; whole model including auxiliary heads = 14,805,191.